

RISE IN × STELLAR PRO HACKATHON 2026 · GENESIS TRACK

payper submission answers

Every question Rise In asks, answered in English and Turkish, with the technical detail and the stack behind each answer. Figures are read from the deployment rather than quoted from memory.

payper.live

github.com/cansarihan/payper

Stellar testnet · protocol 28

Can Sarıhan · Berk Çiçek

1 · The Narrative "Why"

Five questions, asked verbatim in the submission form.

Q1 What are you building?

On-chain receivable financing. A supplier uploads a signed e-invoice; the contract writes its unique document identifier and refuses a second registration of the same one. The buyer acknowledges the debt on chain. From that moment the receivable is an instrument anyone holding USDC can fund — in seconds, from fifty dollars up — and the supplier is paid the same day in their own currency through a SEP-6 anchor.

TR Zincir üstü alacak finansmanı. Tedarikçi imzalı e-faturasını yüklüyor; kontrat faturanın tekil kimliğini yazıyor ve aynısının ikinci kez kaydedilmesini reddediyor. Alıcı borcu zincir üstünde onaylıyor. O andan itibaren alacak, USDC tutan herkesin saniyeler içinde ve elli dolardan başlayarak fonlayabildiği bir enstrüman oluyor; tedarikçiye aynı gün kendi para biriminde, SEP-6 anchor üzerinden ödeniyor.

Q2 What problem does it solve?

A supplier waits 60 to 120 days to be paid while the goods have shipped and the wages are paid. The capital that could bridge that gap is bank capital: a factor can only advance what its own balance sheet allows, and that balance sheet is bounded by the country it is licensed in. A funder abroad who would take that risk at that price cannot reach it — no local-currency account, no correspondent banking, no ticket size small enough to be worth the paperwork. Two more problems follow: the discount is asserted rather than derived, and the same receivable can be sold twice.

TR Tedarikçi malı sevk etmiş, maaşı ödemiş; parayı 60-120 gün bekliyor. O açığı kapatacak sermaye banka sermayesi: bir faktoring şirketi ancak kendi bilançosunun izin verdiği kadar avans verebiliyor ve o bilanço, lisanslı olduğu ülkeyle sınırlı. Bu riski o fiyata almaya razı yabancı bir fonlayıcı ise erişemiyor — yerel para hesabı yok, muhabir bankacılık yok, evrak masrafına değecek kadar küçük bir dilim yok. İki sorun daha bunu izliyor: iskonto türetilmiyor, dayatılıyor; ve aynı alacak iki kez satılabilir.

Q3 Who are your target users?

Suppliers invoicing a single large corporate buyer — organised retail, automotive sub-industry, construction materials. Ticket sizes of roughly 1,500 to 15,000 USD equivalent on 30 to 120 day terms. On the other side, anyone holding USDC who wants short-dated, buyer-acknowledged paper. The buyer's acknowledgement is both the lock that makes the receivable real to a funder and the distribution channel: one corporate buyer brings its suppliers with it.

TR Tek büyük kurumsal alıcıya fatura kesen tedarikçiler — organize perakende, otomotiv yan sanayi, inşaat malzemesi. 30-120 gün vadede, yaklaşık 1.500-15.000 USD karşılığı dilimler. Diğer tarafta, kısa vadeli ve alıcısı onaylı kâğıt arayan, USDC tutan herkes. Alıcının onayı hem alacağı fonlayıcı gözünde gerçek kılan kilit hem de dağıtım kanalı: bir kurumsal alıcı, tedarikçilerini beraberinde getiriyor.

Q4 Why is this problem worth solving?

The market is established rather than speculative — it already pays. In the first corridor the figures are public: Türkiye had 4,016,059 active enterprises in 2025, of which 99.6% are SMEs (TÜİK), and the factoring sector turned over 1.875 trillion TRY across roughly 95,000 customers in 2025. What that market does not have is a price anyone can check, or capital from outside the licensing perimeter. Both are addressable.

TR Pazar spekülatif değil, kurulu — ve zaten ödüyor. İlk koridorda rakamlar açık: Türkiye'de 2025'te 4.016.059 aktif girişim vardı, bunların %99,6'sı KOBİ (TÜİK); faktoring sektörü 2025'te yaklaşık 95.000 müşteri üzerinden 1,875 trilyon TL hacim döndürdü. Bu pazarda olmayan şey, herkesin kontrol edebileceği bir fiyat ve lisans çevresinin dışından gelen sermaye. İkisi de çözülebilir.

Q5 What is your value proposition?

For whom	What changes
Supplier	Paid the same day, in local currency, into a normal bank account. Never touches crypto.
Funder	Reaches a market that was closed, from fifty dollars, without opening an account in that country.
Buyer	Suppliers stay liquid without the buyer extending credit or paying early.
Everyone	The discount is computed from on-chain data, each component labelled with its source.

TR Tedarikçi aynı gün, kendi para biriminde, normal banka hesabına ödeme alıyor ve kriptoya hiç dokunmuyor. Fonlayıcı kapalı bir pazara elli dolardan, o ülkede hesap açmadan erişiyor. Alıcı, kendi kredisini uzatmadan tedarikçisini likit tutuyor. Herkes için: iskonto zincir üstü veriden hesaplanıyor ve her bileşen kaynağıyla etiketli.

THE MECHANISM IS NOT COUNTRY-BOUND

The shape of the problem is the same wherever suppliers invoice on terms. What changes per country is the document standard, the invoice identifier and the anchor. Türkiye is the first corridor and what runs on testnet today.

TR · Sorunun biçimi, tedarikçilerin vadeli fatura kestiği her yerde aynı. Ülkeye göre değişen şey belge standardı, fatura kimliği ve anchor. Türkiye ilk koridor ve bugün testnet'te çalışan da o.

2 · MVP · required deliverables

Asked for	Where it is
Public GitHub repository	<code>github.com/cansarihan/payper</code> — public, two co-authors
Well-structured README	Opens with a five-minute evaluation section; architecture, design decisions, technical challenges and a dedicated Honest limitations section
Smart contracts on Stellar Testnet	Four contracts, Soroban SDK 27.0.6, <code>wasm32v1-none</code> , protocol 28
Front-end / application URL	<code>payper.live</code> — our own server behind a Cloudflare Tunnel
Working live demo	A judge can sign in, upload an invoice, price it, fund it and settle it. Every step writes a real transaction to testnet
Documented contract IDs	README Deployed artifacts table, every entry linked to <code>stellar.expert</code>

TR İstenen altı kalemin altısı da yerinde: açık depo, beş dakikada değerlendirme bölümüyle başlayan README, testnet'te dört kontrat, kendi sunucumuzda canlı uygulama, jürinin uçtan uca çalıştırabildiği demo ve `stellar.expert` linkli kontrat adresleri tablosu.

Deployed artifacts

Contract	Address
Invoice contract	<code>CB2EUFAFCDKWHCYBH6DTFN0HJEVYH3WKTKL40TGX5FUKP272GHQ63NBA</code>
Treasury adapter	<code>CAFZQFGPDEHV62AMP6NEMYHV6MEPAKFFPVCXDVUAQJA5Q4QWEFBIA3X4</code>
DeFindex vault	<code>CBXHLM65LG05400WBCIQKRVHJGQPSG6D2J5QSALXKYHJHR2J5RP0DCP</code>
Blend v2 pool · rate reference	<code>CCEBVDYM32YNYCVNRXQKDFFPISJJCV557CDZEIRBEE4NCV4KHPQ44HGF</code>
Local-currency feed (SEP-40)	<code>CC06YMLR2MUB4JYZIU77XC07DP6EVN00AF4ZZQXZOW7UEVNG52XJLRC</code>
Treasury, local fallback	<code>CACRTTWHUUD7KCIJWVYCKJIH6ZM5K2WWHXKWNCCG4PR3X52ALG3NTGDI</code>
USDC (SAC)	<code>CBIELTK6YBZJU5UP2WWQEUCYKLP06AUNZ2BQ4WWFEIE3USCIHMXQDAMA</code>
Anchor	<code>tr-mock-anchor.fly.dev</code>

THE DEPLOYED CONTRACT IS A REPRODUCIBLE BUILD

`.github/workflows/release.yml` runs `stellar.expert`'s build workflow on a tag. What is deployed is that artifact, not a local build — the hashes are identical, and GitHub holds a signed attestation tying the hash to this repository at tag `v1.0.2`. Checkable in three commands that do not trust us.

TR · Zincirdeki `wasm`, `Cl`'in temiz bir runner'da ürettiği artefaktın aynısı — hash'ler birebir eşleşiyor ve GitHub bu hash'i depoya bağlayan imzalı bir attestation tutuyor. Bize güvenmeyi gerektirmeyen üç komutla doğrulanabiliyor.

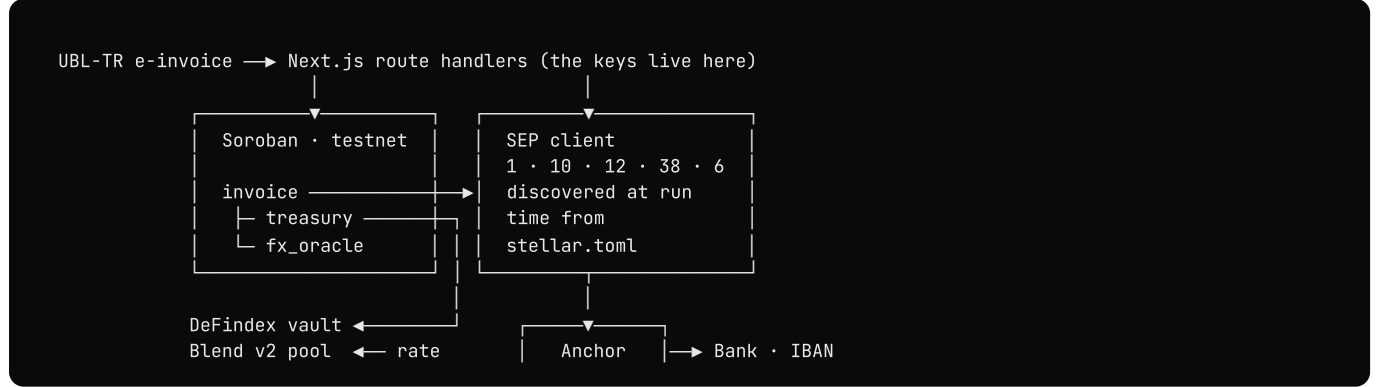
Numbers, read from the deployment

Metric	Value
Rust · contracts	2,744 lines across 10 files
TypeScript · app	17,537 lines across 90 files
Contract tests	36 , all passing
Wallet signing	Verified end to end — prepare, sign outside the server, submit, read back
Application-level checks	100 , across five suites
API routes	22
Panel screens	15
SEP standards implemented	9

3 · Technical documentation

The five items the submission asks to be explained.

T1 Overall architecture



The client holds no chain logic. It renders `/api/state` and posts intents; every signature and every contract call happens in a route handler. That is a deliberate trade-off, listed under design decisions below.

TR İstemcide zincir mantığı yok. Sadece `/api/state`'i render ediyor ve niyet gönderiyor; her imza ve her kontrat çağrısı route handler'da gerçekleşiyor. Bu bilinçli bir ödünleşme, aşağıda yazılı.

T2 Main components and their responsibilities

Component	Responsibility
contracts/invoice	The ETTN registry and the product's one invariant. Registration, buyer acknowledgement, pricing, the funding ledger, transferable claims, repayment and the recourse waterfall.
contracts/treasury_defindex	Holds the pooled capital as DeFindex vault shares. Reports the yield: the vault's own realised gain when it has one, otherwise a Blend v2 pool's lending rate.
contracts/treasury_local	A fallback treasury with the same interface, so the backing vault can fail or be swapped without changing the flow.
contracts/fx_oracle	SEP-40 shaped price feed for the local-currency leg, with the history the volatility estimate reads.
src/lib/ubl	Document verification: mandatory UBL fields, the XAdES signature block's structure, the document hash, and whether the identifier is free.
src/lib/anchor	The SEP client. Discovery, authentication, KYC, firm quotes, deposit and withdrawal, plus the tranche splitter that fits transfers inside the anchor's limits.
src/lib/auth	Passkey registration and login, wallet login by signed message, sessions and role gates.
src/lib/chirp	The audio payment frame: 8-FSK encoder, FFT decoder, CRC-8.

TR Fatura kontratı ETTN kaydını ve ürünün tek değişmezini tutuyor. Hazine adaptörü havuzlanan sermayeyi DeFindex kasa payı olarak tutuyor ve getiriye raporluyor. Yerel hazine aynı arayüzle yedek. Oracle kur bacağını besliyor. UBL katmanı belgeyi doğruluyor, anchor katmanı SEP akışlarını yürütüyor, auth katmanı kimlik ve rolleri, chirp katmanı sesli ödeme çerçevesini.

T3 Stellar integrations and protocols used

Integration	How it is used
Soroban SDK 27.0.6	Four contracts, <code>wasmi32v1</code> - none, protocol 28. All three storage layers: <code>instance</code> for config, <code>persistent</code> for the ETTN registry with TTL extended to the maximum, <code>temporary</code> for the quote lock, which expires by itself.
DeFindex	The treasury is a vault opened through their own factory. A contribution becomes vault shares; a payout burns the shares it is worth. Our vault's wasm hash matches DeFindex's published hash byte for byte.
Blend v2	The yield component reads the pool's <code>bToken</code> index, sampled once when the reference is set and annualised over a window the adapter times itself. Live: 127 bps.
Stellar Wallets Kit 2.6	Wallet connection for funders and suppliers who bring their own key.
SEP-1	Anchor discovery. Every endpoint is read at run time, nothing hard-coded.
SEP-6	Programmatic deposit and withdrawal, both directions.
SEP-7	The payment URI in the QR code, so any wallet can read what the tones carry.
SEP-10	Challenge authentication, verified before signing.
SEP-12	Customer registration for the fiat leg.
SEP-38	Firm quotes, so the payout shown is the payout paid.
SEP-40	The oracle interface the currency feed is read through.
SEP-41	The treasury position itself is a SEP-41 token — the DeFindex vault share.
SEP-46	Contract metadata in the wasm: <code>source_repo</code> and <code>home_domain</code> , readable straight off the ledger.
SEP-53	Signed messages for wallet login, verified against the spec's own test vectors.

WHICH INTEGRATION IS LOAD-BEARING

Without the anchor the supplier cannot be paid and the buyer cannot settle — that is the product's entire proposition. Without the treasury, `fund()` cannot complete and the yield component of the price has no source. Neither is decoration.

TR · Anchor olmadan tedarikçi ödenemiyor ve alıcı kapatamıyor — ürünün bütün önerisi bu. Hazine olmadan `fund()` tamamlanamıyor ve fiyatın getiri bileşeninin kaynağı kalmıyor. İkisi de süs değil.

T4 Key design decisions and trade-offs

Decision	What it costs, and why we took it
The identifier is the invariant	Uniqueness is a contract rule, not a database row. A licensed factor inside a country with a registry can already check one; a funder abroad cannot. The contract gives them the same guarantee without asking them to join anything.
Every price component carries its source	More surface to build and to explain, but a number that fell back to a parameter cannot arrive wearing the live badge. This is the decision the rest of the product is organised around.
SEP-6 rather than SEP-24	SEP-24 hands the flow to the anchor's own hosted interface. SEP-6 is API-first, which is what lets us show every request and status in the open. The cost is that we implement the flow ourselves.
The treasury is an interface, not a vault	The invoice contract knows four functions and cannot tell which treasury is behind them. That is why a DeFindex outage or a swap does not touch the flow.
Yield falls through, never substitutes	The vault's realised gain first, a Blend pool's lending rate second, and a refusal third. It would have been easier to return the configured number and call it live.
Who signs depends on how you arrived	A connected wallet signs its own transactions: the server prepares and simulates the call, hands over an unsigned envelope, and the extension shows what it authorises. Demo and passkey sessions are signed server-side — three parties on one laptop makes the demo legible, and a passkey produces secp256r1 where Stellar expects Ed25519, so it cannot sign Soroban XDR until a smart account verifies that curve on chain.

TR Kimliğin değişmez olması, her fiyat bileşeninin kaynağını taşıması, SEP-24 yerine SEP-6, hazinenin kasa değil arayüz olması, getirinin sayı uydurmak yerine reddetmesi, ve anahtarların sunucuda durması. Sonuncusu gerçek bir maliyet ve sınırlılıklar bölümünde açıkça yazılı — üretimde ilk değişmesi gereken şey bu.

T5 Technical challenges and how we solved them

Challenge	Resolution
Cross-contract authorisation	The treasury moves the token one frame deeper than the invoker's signature reaches. <code>authorize_as_current_contract()</code> authorises that sub-call only, for that amount only.
The DeFindex vault pulls with transfer, not <code>transfer_from</code>	The first adapter granted an allowance and came back with <code>Auth, InvalidAction</code> . We authorise the vault's pull explicitly with the same pattern. Deposit and withdraw verified on testnet with real funds.
The anchor names its memo type	We sent <code>MEMO_TEXT</code> where it asked for <code>MEMO_ID</code> . The payment arrived at the treasury and the transfer sat at <code>pending_user_transfer_start</code> for ever. A simulated rail would not have cared — this bug is the proof the rail is real.
Anchor transfer limits	3,000 ₪ per transfer, 50 ₪ minimum. The splitter broke twice: greedy filling left a remainder under the minimum, even division piled it onto the last part and crossed the cap. The version that rounds parts up and leaves the shortfall last was swept across 1,823 amounts.
Asset naming differs between SEP-38 and SEP-6	SEP-38 wants <code>stellar:USDC:ISSUER</code> and <code>iso4217:TRY</code> ; SEP-6's exchange endpoints want the bare code for the asset but the ISO form for fiat. Found through three separate 400s.
An hour of lending rounds to nothing	Dividing to basis points before scaling to a year truncated the whole measurement to zero, and the yield reading silently disappeared. <code>annualise_bps</code> multiplies first; four tests hold it against two real readings of the pool taken eighty-five minutes apart.
An unbounded funding ledger	Found by running the official Stellar security checklist. <code>repay()</code> walks the ledger and <code>transfer_claim()</code> rebuilds it, so enough entries would strand an invoice. Closed with a minimum ticket and a cap of 32 entries, held by three tests.
Audio on iOS	Four rounds: a suspended <code>AudioContext</code> , a carrier above the voice-band roll-off, a preamble that recurred inside payloads, and a room's reverb merging bursts. The context is now created before the first <code>await</code> , and the detection threshold adapts to a measured noise floor.
Next.js route handler isolation	Route handlers are bundled separately and share no module state. Challenges, the rate limiter and the session secret live on <code>globalThis</code> ; payout accounts and passkeys are written to disk atomically, so a deploy does not reset the demo.

TR Kontratlar arası yetkilendirme, DeFindex'in transfer kullanması, anchor'ın memo tipi, transfer limitleri, SEP-38 ile SEP-6 arasındaki varlık adlandırma farkı, bir saatlik getirinin sıfıra yuvarlanması, sınırsız fonlama defteri, iOS'ta ses ve route handler yalıtımı. Dokuzunun da çözümü yukarıda; memo tipi hatası özellikle önemli, çünkü **simüle bir ray bunu umursamazdı**.

4 · Technology stack

Contracts

Rust soroban-sdk 27.0.6 wasm32v1-none protocol 28 stellar-cli 27 cargo test

Application

TypeScript 6 Next.js 15 · App Router React 19 @stellar/stellar-sdk 17 tsx

Stellar ecosystem

DeFindex Blend v2 Stellar Wallets Kit 2.6 SEP-6 anchor Horizon Soroban RPC stellar.expert

Identity

@simplewebauthn 14 WebAuthn · passkeys SEP-53 signed messages Ed25519

Documents and signals

fast-xml-parser UBL 2.1 / UBL-TR XAdES structure SHA-256 Web Audio API · 8-FSK CRC-8 qrcode

ISO 7064 mod-97 IBAN

Infrastructure

Debian 13 systemd Cloudflare Tunnel GitHub Actions stellar-expert build workflow Sigstore attestations

TR Kontratlar Rust ve soroban-sdk 27.0.6 ile, protokol 28 hedefli. Uygulama TypeScript, Next.js 15 ve React 19. Ekosistem tarafında DeFindex, Blend v2, Stellar Wallets Kit ve SEP-6 anchor. Kimlik WebAuthn passkey ve SEP-53 imzalı mesaj. Belge doğrulama UBL 2.1/UBL-TR ve XAdES yapısı üzerinden. Altyapı kendi sunucumuzda, derleme GitHub Actions ile yeniden üretilir.

5 · The three track requirements

Requirement	How it is met
Integration from the eligible partners list	DeFindex (DeFi-Yield) holds the treasury; Blend v2 (DeFi-Lending) supplies the yield rate; Stellar Wallets Kit (Wallets) connects funders. Three entries from the list.
Anchor / Local Payments a real fiat rail	SEP-6 in both directions, discovered at run time from <code>stellar.toml</code> . The smoke run has two funders buy USDC through the anchor before funding, and the supplier sells the payout back for local currency.
Core Feature load-bearing, not an add-on	Remove the anchor and the supplier cannot be paid. Remove the treasury and <code>fund()</code> cannot complete. Neither can be deleted without deleting the product.

TR Üç zorunlu şartın üçü de karşılanıyor: uygun listeden üç entegrasyon, çift yönlü SEP-6 fiat rayı, ve ikisinin de taşıyıcı olması — çıkarıldıklarında ürün çalışmıyor.

6 · Stellar skill files used

The submission asks which official skill files were used, by path. These, from `stellar/stellar-dev-skill`:

Path	What it was used for
skills/standards/SKILL.md	The SEP/CAP routing map. Confirmed SEP-6 over SEP-24 for an API-first fiat rail, and SEP-12 for the KYC leg. SEP-46 and SEP-55 came from here.
skills/smart-contracts/SKILL.md	Build and platform constraints. Our release profile, <code>crate-type</code> , target and event usage were checked line by line and already matched.
skills/smart-contracts/security.md	The contract security checklist, run against <code>contracts/invoice</code> .

WHAT THE SECURITY REVIEW FOUND — AND WE FIXED

Every privileged entry point authorises from stored config, `init` refuses a second call, storage keys are typed, arithmetic is checked, error codes are appended rather than renumbered, and events cover state changes. One item did not pass: *"loops and footprints bounded on attacker-shaped input."* `fund()` accepted any amount above zero and appended to an unbounded ledger that `repay()` walks. Closed with a configurable minimum ticket — waived only for the contribution that closes an invoice, so the floor can never strand one — and a cap of 32 entries. Three tests hold it.

TR · Resmî güvenlik listesini kontratımıza uyguladık. Bir madde geçmedi: `fund()` sıfırdan büyük her tutarı kabul ediyor ve `repay()`'in dolaştığı sınırsız bir deftere ekliyordu. Asgari dilim ve 32 kayıt üst sınırıyla kapatıldı, üç testle bağlandı.

7 · The judging criteria, answered

J1 Meaningful idea & real-world impact

The problem is defined by a number a supplier can read off their own ageing report: sixty to a hundred and twenty days. The use case is trade finance and local payments, both named in the brief. Who benefits: suppliers who stay solvent without a bank line, and funders who reach a market that was closed to them. The target segment is specific — suppliers invoicing one large corporate buyer, 1,500 to 15,000 USD per invoice — which is also the distribution channel, because one buyer brings its suppliers.

TR Problem, tedarikçinin kendi vade raporundan okuyabileceği bir sayıyla tanımlı: 60-120 gün. Kullanım alanı ticaret finansmanı ve yerel ödemeler, ikisi de şartnamede geçiyor. Kazanan: banka limiti olmadan ayakta kalan tedarikçi ve kendisine kapalı bir pazara erişen fonlayıcı. Hedef kitle belirli ve aynı zamanda dağıtım kanalı.

J2 Technical implementation

What is asked	Evidence
Deployed with real functionality, not mocked	Four contracts on testnet. <code>npm run smoke</code> registers, acknowledges, prices, funds from two funders who buy USDC through the anchor first, settles, distributes pro rata, then tries the same identifier again and is refused. Every step a real transaction.
Core flows work end to end	The smoke run is that flow, and it is also driveable by hand in the panel.
Soroban auth and storage used appropriately	All three storage layers, each for what it is for. Cross-contract authorisation scoped to one sub-call and one amount.
Architecture designed and documented	A rendered diagram in the README plus a component table, design decisions and challenges.
Passkeys and smart wallets are a bonus	Implemented — passkey registration and login, and SEP-53 wallet login verified against the spec's vectors. A connected wallet also signs its own transactions; <code>npm run test:wallet</code> asserts the envelope the server hands over carries no signature of its own.

J3 Ecosystem fit – anchor carries the highest weight

Three integrations from the eligible list, all load-bearing. The anchor is SEP-6 in both directions and discovered at run time, so moving to another anchor is one domain change. The DeFindex claim is cryptographic rather than narrative: our vault's wasm hash matches their published hash byte for byte. The skill files are cited by path in the README, and the security checklist from one of them found a real bug that is now fixed.

TR Uygun listeden üç entegrasyon, hepsi taşıyıcı. Anchor çift yönlü SEP-6 ve çalışma anında keşfediliyor — başka bir anchor'a geçmek tek alan adı değişikliği. DeFindex iddiası anlatı değil kriptografik: kasamızın wasm hash'i yayımladıklarıyla bayt bayt aynı. Skill dosyaları README'de yol vererek atıflı.

J4 User experience

The interface is built for someone who has never held a token: the supplier signs in with Face ID, never sees a seed phrase, and is paid in their own currency into a normal bank account. Fifteen screens, each one a step with its own number, so the order is discoverable without instruction. A guided tour walks the whole flow. English is the default; Turkish is one toggle away.

TR Arayüz, hiç token tutmamış biri için kurulu: tedarikçi Face ID ile giriyor, seed phrase görmüyor, kendi para biriminde normal banka hesabına ödeme alıyor. On beş ekran, her biri numaralı bir adım, böylece sıra anlatılmadan anlaşılıyor. Rehberli tur bütün akışı geziyor. Varsayılan dil İngilizce, Türkçe tek tuş.

J5 Traction & continuity

Feedback during the event changed the product twice. A jury member asked what the blockchain part actually does; we reframed from digitising factoring to opening the capital pool across borders, and rewrote the site and README around it. A

judge said a DeFi yield integration would put us in finalist territory; the Blend v2 reading exists because of that conversation, and the yield component reads live at 127 bps as a result. Both changes are visible commit by commit in the git history. The roadmap and the intended next step — SCF and InstaAward — are in the README.

TR Etkinlikteki geri bildirim ürünü iki kez değiştirdi. Bir jüri üyesi blockchain'in gerçekte ne yaptığını sordu; konumlandırmayı faktoringi dijitalleştirmekten sermaye havuzunu sınır ötesine açmaya çevirdik ve siteyi de README'yi de buna göre yeniden yazdık. Bir jüri DeFi getiri entegrasyonunun finale taşıyacağını söyledi; Blend v2 okuması tam olarak o konuşmadan çıktı ve getiri şu an 127 bps olarak canlı okunuyor. İki değişiklik de git geçmişinde commit commit görünüyor.

36 Presentation & documentation

The README opens with a five-minute evaluation path and a table of what to look at, in order. Setup, testing and evaluation commands are copy-pasteable and were run as written. Architecture is a rendered diagram. The proof section gives commands that do not require trusting us. Open limitations have their own section rather than being buried.

TR README beş dakikada değerlendirme yolu ve sırayla nereye bakılacağı tablosuyla açılıyor. Kurulum, test ve değerlendirme komutları kopyala-yapıştır ve yazıldığı gibi çalıştırıldı. Mimari render edilmiş bir diyagram. Kanıt bölümü bize güvenmeyi gerektirmeyen komutlar veriyor. Açık sınırlılıklar gömülü değil, kendi bölümünde.

8 · Honest limitations

STATED HERE RATHER THAN FOUND LATER

Nobody bears the currency risk yet. The funder's claim is fixed in USDC while the buyer owes a fixed amount of local currency. Over ninety days those stop matching, and the contract does not say who absorbs the difference. The currency premium prices that risk; it does not assign it.

The funder's claim has no legal wrapper. The contract creates a pro-rata economic interest in a payout, not an assignment of the receivable. What that is under securities law differs by country.

A fabricated invoice passes our checks. We verify structure, not existence. The gate is `acknowledge()` — the address written on the invoice must sign. What that does not stop is collusion between a supplier and a buyer.

Also: the signature is checked structurally, not against the tax authority's certificate chain; the bank leg runs against a sandbox anchor; and we publish the testnet price feed ourselves. A connected wallet signs its own transactions; demo and passkey sessions are still signed server-side, and for passkeys that stays open until `secp256r1` is verified on chain.

TR · Kur riskini henüz kimse taşıyor. Fonlayıcının hakkının hukuki kabı yok. Uydurma fatura yapısal kontrollerden geçer — kapı `acknowledge()`, ama muvazaayı durdurmuyor. Ayrıca imza GİB sertifika zincirine karşı doğrulanmıyor, banka bacağı sandbox, kur beslemesini testnet'te biz yayımlıyoruz ve fonlama işlemleri — bağlı bir cüzdan kendi işlemini imzalıyor, ancak demo ve passkey oturumları hâlâ sunucuda imzalanıyor.

9 · After the hackathon

Next step	Scope
Stellar Community Fund	The missing pieces are a passkey smart account that verifies <code>secp256r1</code> on chain, so a passkey signs its own transactions, and integration with a production fiat anchor. Both are scoped work.
InstaAwards	The passkey smart account is a well-sized scope on its own.
Closed pilot	One corporate buyer and the suppliers that invoice it. The buyer's acknowledgement is the product's lock, so the pilot starts there.
DeFindex strategies	Attaching a strategy to the vault turns the treasury from custody into yield, and the adapter already prefers the vault's own gain when it has one.

TR SCF için eksik parçalar: secp256r1'i zincirde doğrulayan passkey akıllı hesabı ve üretim anchor'ı entegrasyonu. InstaAward için passkey akıllı hesabı tek başına yeterli kapsam. Pilot bir kurumsal alıcı ve tedarikçileriyle başlıyor. Kasaya strateji bağlanması hazineyi saklamadan getiriye çeviriyor.

TEAM

Can Sarihan — anchor and SEP client, identity and sessions, the API surface, panel screens.

Berk Çiçek — contracts, the document validator, README and technical documentation.

Component ownership rather than a split of hours: whoever is named owns how that part works.